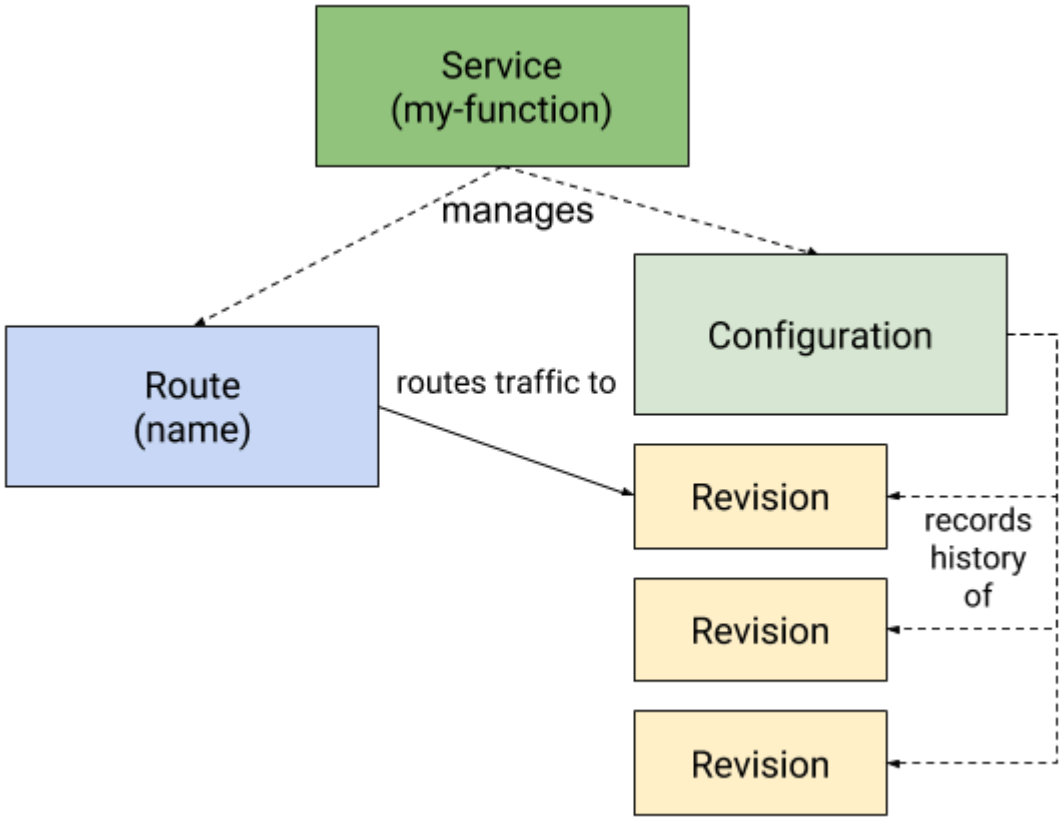
Knative Serving

Knative Serving defines a set of objects as Kubernetes Custom Resource Definitions (CRDs). These resources are used to define and control how your serverless workload behaves on the cluster.



The primary Knative Serving resources are Services, Routes, Configurations, and Revisions:

- **Services**: The `service.serving.knative.dev` resource automatically manages the whole lifecycle of your workload. It controls the creation of other objects to ensure that your app has a route, a configuration, and a new revision for each update of the service. Service can be defined to always route traffic to the latest revision or to a pinned revision.
- **Routes**: The `route.serving.knative.dev` resource maps a network endpoint to one or more revisions. You can manage the traffic in several ways, including fractional traffic and named routes.
- **Configurations**: The `configuration.serving.knative.dev` resource maintains the desired state for your deployment. It provides a clean separation between code and

configuration and follows the Twelve-Factor App methodology. Modifying a configuration creates a new revision.

- **Revisions**: The `revision.serving.knative.dev` resource is a point-in-time snapshot of the code and configuration for each modification made to the workload. Revisions are immutable objects and can be retained for as long as useful. Knative Serving Revisions can be automatically scaled up and down according to incoming traffic.

For more information on the resources and their interactions, see the **Resource Types Overview** in the `serving` Github repository.

Common use cases

Examples of supported Knative Serving use cases:

- Rapid deployment of serverless containers.
- Autoscaling, including scaling pods down to zero.
- Support for multiple networking layers, such as Contour, Kourier, and Istio, for integration into existing environments.

Knative Serving supports both HTTP and **HTTPS** networking protocols.

Installation

You can install Knative Serving via the methods listed on the **installation page**.

Getting Started

To get started with Serving, check out one of the **hello world** sample projects. These projects use the `Service` resource, which manages all the details for you.

With the `Service` resource, a deployed service will automatically have a matching route and configuration created. Each time the `Service` is updated, a new revision is created.

More samples and demos

- **Knative Serving code samples**

Debugging Knative Serving issues

- **Debugging application issues**

Configuration and Networking

- **Configuring cluster local routes**
- **Using a custom domain**
- **Traffic management**

Observability

- **Serving Metrics API**

Known Issues

See the **Knative Serving Issues** page for a full list of known issues.